



The cost of computation and supercomputers

FLOPs, GPU hours, dollars and energy

Carlos Cotrini



The same three questions, asked for the AI

Was kann *es* wissen? → Weekend 5: RAG

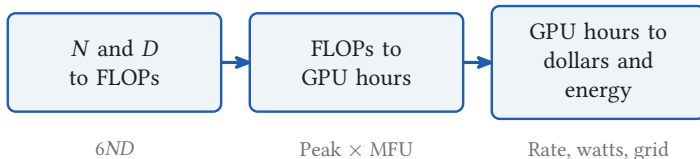
Was soll *es* tun? → Weekend 6: Agentic AI

Was darf *es* hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

Weekend 5 was what it can know, Weekend 6 what it should do. And this weekend is what it may become.



What the next hour computes





Section 1

Dividing a computation, 1793



Adam Smith, 1776



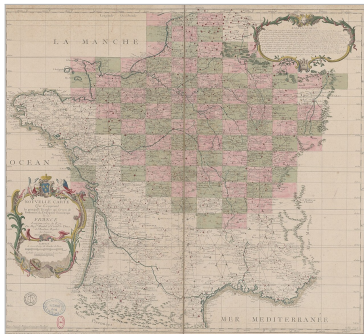
The pin maker's workshop. *Epinglier*, Diderot and d'Alembert's

Encyclopedie, 1762

One man draws out the wire, another straightens it, a third cuts it, a fourth points it, a fifth grinds it at the top for receiving the head.

- *The Wealth of Nations*, Book I, chapter 1: **Of the Division of Labour**
- One workman making a whole pin: perhaps twenty in a day, perhaps not one
- Ten men, split across eighteen operations: **forty eight thousand**
- Same people, same tools. Only the organisation changed

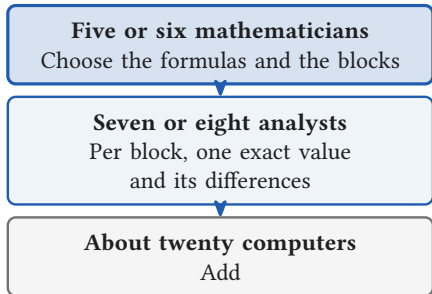
The task: the area of every parcel in France



The principal triangles of France. Maraldi and Cassini de Thury, 1744

- The Revolution replaces tax privilege with one tax on land
- So every parcel has to be measured, and its area computed
- Gaspard de Prony directs the Bureau du Cadastre, which does the computing
- Land is measured in triangles, because angles are quick and chains are slow

Three tiers, and what each hands down

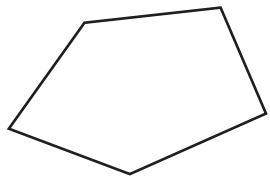


A computing room, Harvard College Observatory, about 1890

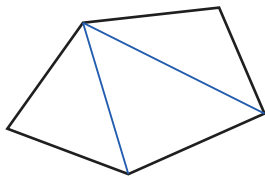
Je conçus tout à coup l'idée d'appliquer la même méthode à l'immense travail dont je m'étais laissé imposer le fardeau, et de fabriquer mes logarithmes comme on fabrique des épingles.

Manufacturing my logarithms as one manufactures pins.
Gaspard de Prony, Notices, 1824, p. 5.

One parcel, many triangles, one each



One parcel



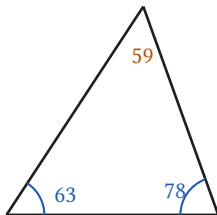
Cut into triangles



One each

- A clerk never sees a parcel. He is handed one triangle and returns one area
- Which is why twenty of them can work at once without talking to each other

One side, two angles, and a table of logarithms



400 m

Measured with the chain

The third angle is free:

$200 - 63 - 78$

$$\log \text{Area} = 2 \log c + \log \sin A + \log \sin B \\ - \log \sin C - \log 2$$

$2 \log 400$	+5.2041200	
+ $\log \sin 63$	-0.0778938	
+ $\log \sin 78$	-0.0264654	
- $\log \sin 59$	-0.0970812	
- $\log 2$	+0.3010300	
$\log \text{Area}$	4.8958120	
Area	78 670 m ²	about 7.9 ha

Five lookups in a seven figure table, then four additions. The parcel is ours; the units and the method are the period's.



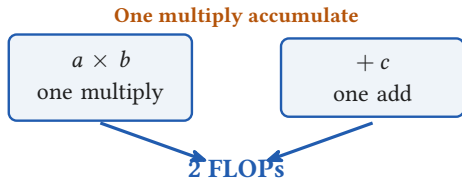
Section 2

One operation, and who does it

2

☰ One operation, and who does it

A FLOP is one operation on two numbers



Count, or rate

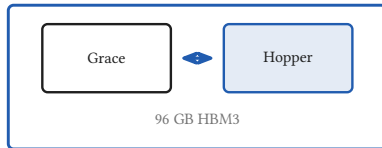
FLOP: a count of operations
how much work

FLOP/s: operations per second
how fast

One computer, then and now



A computer room, NACA, 1949



One GH200. Alps holds 10,752

Human computer, 1949

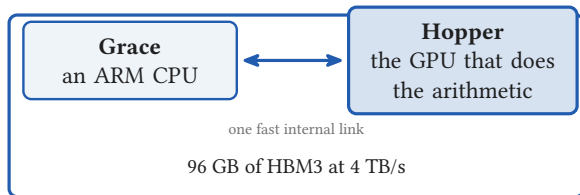
Rate	1,000 additions a day
Format	Decimal, by hand
Power	About 100 W
Food	2,065 kcal a day
	what one person eats

One GH200, 2024

990 TFLOP/s dense BF16
16 bits, fixed
660 W on Alps
13,600 kcal a day
what six or seven people eat

- The clerk's rate is what the shop *delivered*. The chip's is what it *could* do

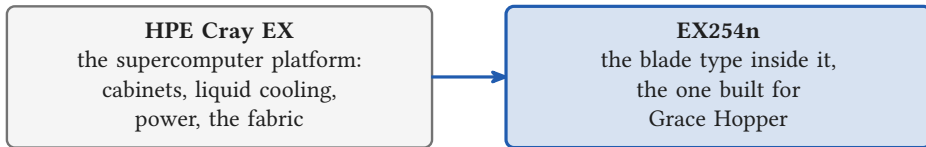
A GH200 is a CPU and a GPU on one module



- The GPU computes. The CPU reads the data, runs Python, and keeps it fed
- HBM3 is memory stacked on the GPU package: only 96 GB, but 4 TB/s
- Between them 900 GB/s, about seven times what a plug-in card gets

Alps is an HPE Cray EX254n

The TOP500 entry reads: *Alps, HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS*



The trailing n marks the NVIDIA accelerated blades

- The blade choice is what makes Alps a machine for AI rather than a conventional CPU cluster

The atelier against Alps

A collection of computers	Rate	Power
Prony's atelier, 1795 twenty people	0.694 additions/s	about 2 kW
Alps, 2024 10,752 modules	10.64 EFLOP/s	7.1 MW

Fifteen billion billion times the rate, on three and a half thousand times the power

What the TOP500 actually measures

A list of the 500 most powerful known supercomputers, published every June and November since 1993. Alps is rank 10 in June 2026.

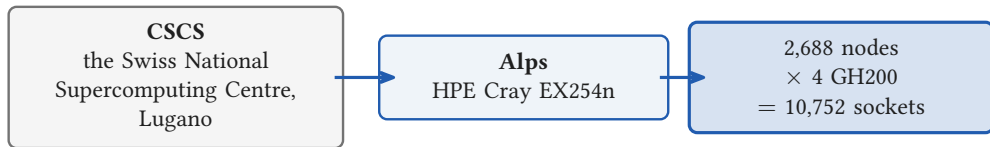
It ranks them on one benchmark: **HPL**, solving a large dense linear system in **64 bit** arithmetic

Rmax 434.90 PFLOP/s
what it actually achieved

Rpeak 574.84 PFLOP/s
the theoretical maximum

So even on its own benchmark it reaches 75.7 percent, and drew 7.1 MW doing it

CSCS and Alps, the machine behind the anchor



At dense BF16 the whole partition is about **10.6 EFLOP/s**, which is the rate that matters for training.

Its TOP500 entry, 434.90 PFLOP/s at 7,124 kW, is an **fp64** measurement and is about 24 times smaller.

Machine: top500.org/system/180259 and cscs.ch/computers/alps.

One second of a GH200, in three registers

The same one second of arithmetic

The mean age of all 8.2 billion people alive,
one addition each

8.3 microseconds

130,000 photographs classified by ResNet-50,
each 224 by 224 pixels, or 0.05 megapixels

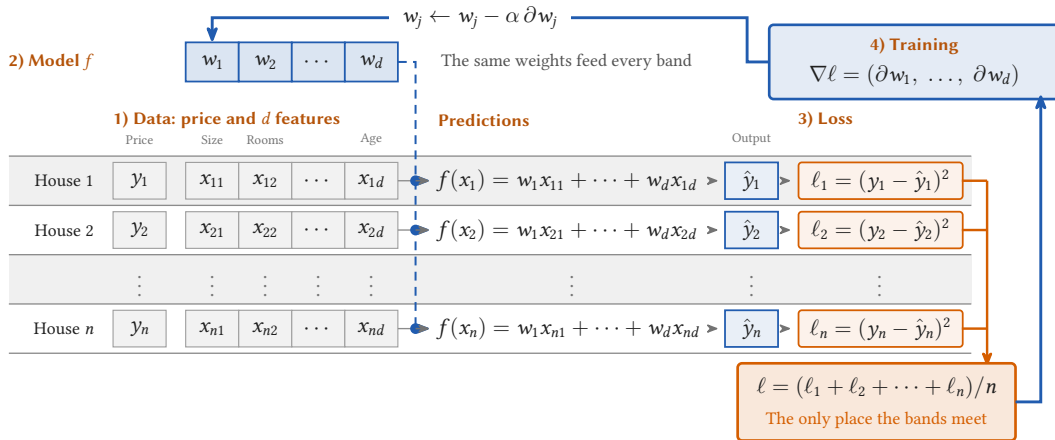
one second

Prony's atelier doing that same
one second of work

45 to 136 million years

ResNet-50: He et al., [arXiv:1512.03385](https://arxiv.org/abs/1512.03385), 3.8 billion multiply-adds, so 7.6 GFLOP per image. A peak count.

One training step of linear regression





Section 3

What the headline rate hides

3

One floating point number, taken apart

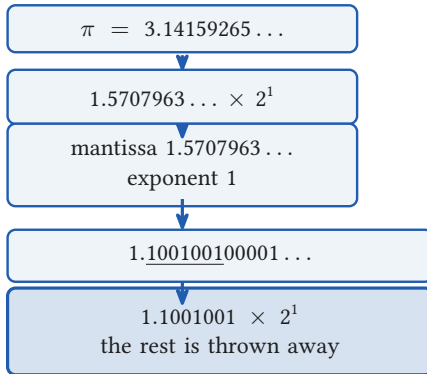
Start with

A mantissa times a power of two

So really two numbers

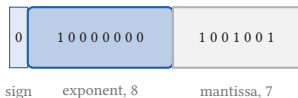
The mantissa in binary

bf16 keeps seven of those bits



What the seven bits cost

The sixteen bits



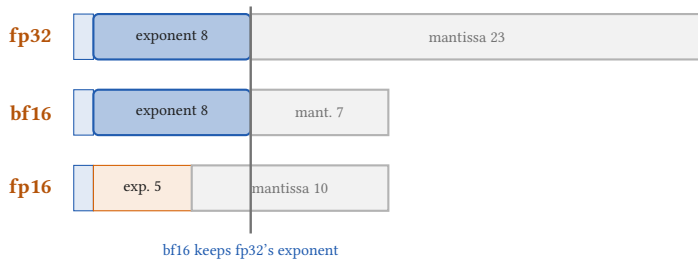
Back to decimal
3.140625

We started from
3.14159265...

Lost 0.00097, or 0.03 percent

- Seven bits cannot hold π , so the number stored is not the number you asked for

Not every FLOP costs the same



Why we say dense BF16, and not just BF16

The datasheet's fast number assumes a pattern



Two of every four pinned to zero, so the hardware skips them

1,979 TFLOP/s

Pretraining cannot promise it



Every weight starts random and has to stay free to move

990 TFLOP/s per GH200, dense

- Sparsity is just something you impose on a finished model

Capacity and bandwidth are different numbers

96 GB
how much fits

Decides whether the weights and the optimiser state fit on one device at all

4 TB/s
how fast you can move it

Decides how quickly the arithmetic units can be fed once the data is there

- 96 GB is less than an ordinary server holds. That is the trade: HBM is stacked on the package and wired short and wide, which is what buys the speed

Two hundred and forty six operations per byte



$$\frac{990 \times 10^{12} \text{ FLOP/s}}{4.023 \times 10^{12} \text{ bytes/s}} = 246 \text{ FLOP per byte}$$

A rate over a rate: the seconds cancel, and what is left is operations per byte

So unless the computation reuses each byte about that many times, the units wait. A large matrix multiply does reuse. Many other operations do not.

- This is one of the reasons a real run reaches 21 to 46 percent of peak

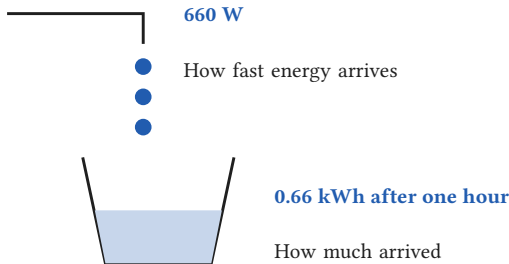


Section 4

The machine, and what it draws

4

A watt is a rate, a kilowatt hour is an amount



Watch the slash

km/h the slash divides,
so it is a speed

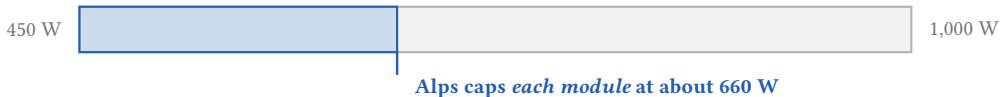
kWh no slash. The hour
multiplies, so it is an amount

Longer gives you *more*, so the hour
cannot be dividing

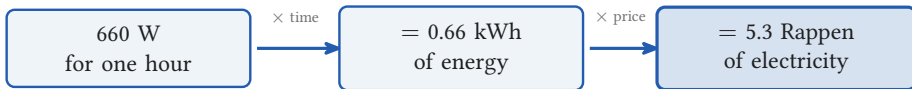
- Rate $\times$ time = amount, twice over: FLOP/s $\times$ seconds = FLOPs, and watts $\times$ hours = kWh

A wattage becomes a bill in two multiplications

One module is just a band



The operator chooses: near 1,000 W is faster, lower saves energy and cooling

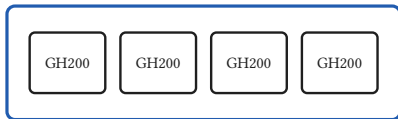


450 to 1,000 W: GH200 datasheet. 660 W: the Alps cap, docs.cscs.ch/running/slurm. 8 Rappen per kWh assumed (ElCom 2026 large consumer band 6.75 to 9; households pay 27.7).
Check: $10,752 \times 660 \text{ W}$ is 7,096 kW against the 7,124 kW TOP500 measured during HPL, top500.org/system/180259.

- That is **one module**. All 10,752 is 7.1 megawatts, about 570 francs an hour
- The whole Apertus run is 3.96 GWh, about 0.3 million francs

Nodes and sockets

One node



Socket one processor module
Node one independent computer

One operating system, one address space

$2,688 \text{ nodes} \times 4 \text{ modules} = 10,752 \text{ sockets}$

Socket to socket, inside the node: 150 GB/s

Between nodes: 25 GB/s per GPU, six times less

- The node boundary is where the bandwidth falls off

Four links, and each step down costs you

- **4,023 GB/s** the GPU to its own HBM3
- **900 GB/s** Grace to Hopper, inside one module
- **150 GB/s** socket to socket, inside a node
- **25 GB/s** per GPU between nodes

Each step down costs about a factor of five. Top to bottom, 161 times.

- So you keep the data at the fast end and cross the slow links as rarely as you can

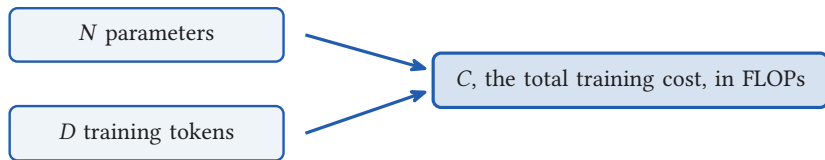


Section 5

From a model to GPU hours

5

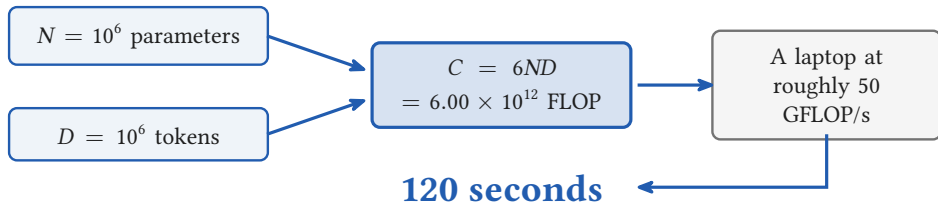
Two numbers price a training run



$$C = \underbrace{6}_{\text{per parameter, per token}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{tokens}}$$

[Kaplan et al. 2020](#), section 2.1. Where the 6 comes from is the 12:00 lecture; for the next fifty minutes it is a rule of thumb.

A million by a million is two minutes



The anchor: Apertus, trained on Alps

Apertus-70B

Released 2 September 2025, Apache 2.0
 70.60×10^9 parameters, 15 trillion tokens
1,811 languages, 40 percent of it not English

Alps, at CSCS Lugano

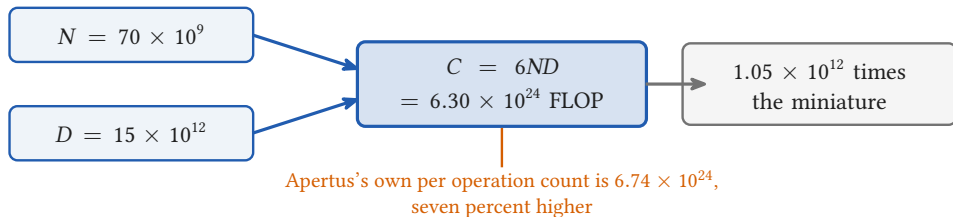
10,752 GH200 sockets, 4 per node
990 TFLOP/s dense BF16 each
TOP500: 434.90 PFlop/s at 7,124 kW

The production run Over 6 million GPU hours, over roughly three months, on up to 4,096 GPUs, sustaining about 723 tokens per second per GPU

Built by EPFL, ETH Zurich and CSCS. Every number in the next forty minutes is about this run.

Model: [arXiv:2509.14233](https://arxiv.org/abs/2509.14233). Run: [arXiv:2604.12973](https://arxiv.org/abs/2604.12973). Machine: top500.org/system/180259.

The same formula on Apertus-70B



From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion tokens, on up to 4,096 GPUs.

4,096 GPUs flat out
for **61 days**

It spanned about 91 days,
so **2,747** on average

The gap is ramp up, restarts and smaller phases: a campaign is not one uninterrupted run at full width

$6 \text{ million GPU hours} \times 660 \text{ W} = \mathbf{3.96 \text{ GWh}} \times 8 \text{ Rappen} = \mathbf{0.32 \text{ million francs}}$



Section 6

Peak, and what a real run gets

6

Counting the workshop is not measuring it



Twenty computers, one desk each

Counted from the doorstep. Nothing was run.

Now watch for an hour

A clerk waits for the next sheet.

Two clerks stop to compare a figure.

The batch is small, so some desks are empty.

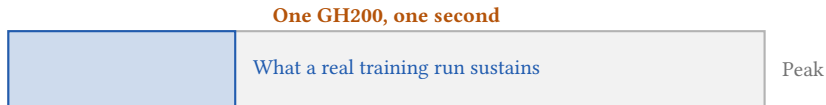
Count the desks,
multiply by how fast
one clerk works



That is the workshop's
peak

- The clerks were at full speed the whole time. They were not always given something to do

Peak is just a ceiling



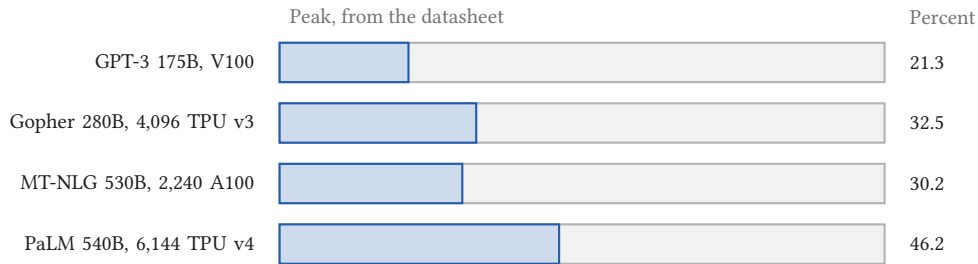
every arithmetic unit busy, every cycle, data always ready

A whole training run, averaged, drawn on one second so you can see it

The gap is waiting on memory, talking to other chips, and kernels that do not fill the hardware.

- Nobody reaches peak. The datasheet number is a ceiling
- The share you do reach is [model FLOPs utilisation](#), written MFU
- It is the blue bar over the whole bar, so MFU 0.30 means thirty percent of peak

Published runs reach 21 to 46 percent of peak



- Two ran on TPUs, so the umbrella word is [accelerator](#): GPU, TPU, or GH200
- Model FLOPs utilisation: the share of peak a training run reaches
- Assuming 100 percent is wrong by a factor of two to five

Where the utilisation formula comes from

$$\text{MFU} = \frac{6ND}{\text{GPU-hours} \times 3,600 \text{ s/h} \times \text{peak FLOP/s}}$$

- Tokens per second is what you observe. FLOP per token is what you count
- Substitute D/T tokens per second, and $6N$ FLOP per token, where T is the wall clock duration of the training run
- T times chips is exactly GPU-seconds, which is what the run reports
- So the measurement enters through the hours, and $6ND$ supplies the work

Six million hours, measured against peak

The consumption

Over 6 million GPU hours, the 70B production run

measured
× datasheet

$$6 \times 10^6 \text{ GPU-h} \times 3,600 \text{ s/h} \times 990 \text{ TFLOP/s per GPU} \\ = 2.14 \times 10^{25} \text{ FLOP of peak capacity}$$

analytic

$$\frac{6.30 \times 10^{24}}{2.14 \times 10^{25}} = 29.5 \text{ percent}$$

6ND for the 70B

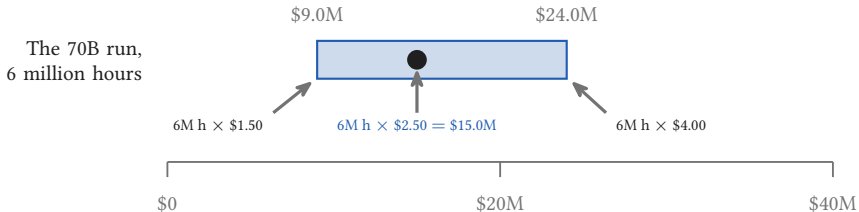


Section 7

Dollars and energy



Six million hours, priced at cloud rates



The \$1.50 to \$4.00 band is an H100 rental survey, November 2025.

The same hours, in gigawatt hours

8.85 million GPU hours
(90 days × 4,096 GPUs)

×

560 W per module

=

4.95 GWh

4.95

3.4



8.6

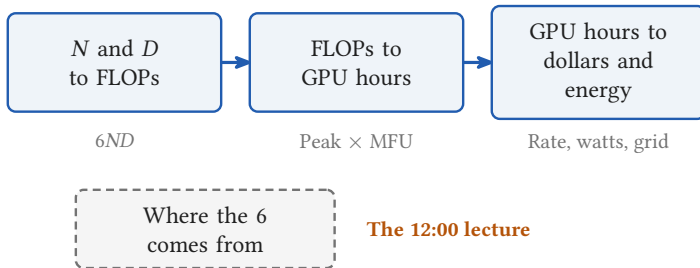


- About 1,100 Swiss households for a year, at 4,500 kWh each

Range 750 to 1,900 households, over four sourced module figures: **560 W** tech report, [arXiv:2509.14233](https://arxiv.org/abs/2509.14233); **660 W** CSCS cap, docs.cscs.ch/running/slurm; **855 W** from the TOP500 run, top500.org/system/180259; datasheet band 450 to 1,000 W.

Household 4,500 kWh a year is ElCom's H4 profile. CSCS publish a PUE below 1.2, cooling from Lake Lugano (cscs.ch).

What you can price at 09:00



- You can now price a training run on the back of an envelope
- At 09:00 you build the envelope